

[813 CHALLENGE // THEME 5 · SUSTAINABLE URBAN PLANNING & SMART CITIES]

CoolCairo

Which blocks should Cairo cool first?

Summer heat risk mapped block by block from satellites, and a 3D tool where planners paint cool roofs, trees, cool pavements and parks and see the effect live.

TEAM 46

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(data & hyperspectral analyst)

EGYPT

Cairo's surfaces pass 45 °C every summer

45.7 °C

a typical east-Cairo block's summer surface temperature (Landsat, 2023–25)

95,141

residents in our 2.2 × 2.2 km Nasr City district (WorldPop 2024)

+1 °C

new blocks built 2016–23 run hotter than established ones (46.9 vs 46.0 °C)

Who is affected: residents of dense blocks on hot surfaces, old and new. Heat hits older people, young children and outdoor workers hardest.

We want to **map** summer heat risk (surface temperature, the materials driving it, and the residents exposed) in **fast-growing MENA cities**, starting with **Nasr City, Cairo**, so that **city planners** can decide which blocks to prioritise for cool roofs, street trees, cool pavements and pocket parks.

End user: a city planner with a limited cooling budget

01 SEE

Where is it hot?

Surface heat and surface materials per 90 m block, in 3D.

02 RANK

Who is exposed?

Heat risk: residents × °C above a typical block.

03 PAINT

Try a measure

Cool roofs, street trees, cool pavements, pocket parks, painted onto blocks.

04 COMPARE

What changes?

Live before → after: temperature, heat exposure, residents cooled.

WHO ADOPTS IT

City and governorate planning departments, new-city developers, and climate-adaptation programmes that fund cooling.

WHY IT PAYS

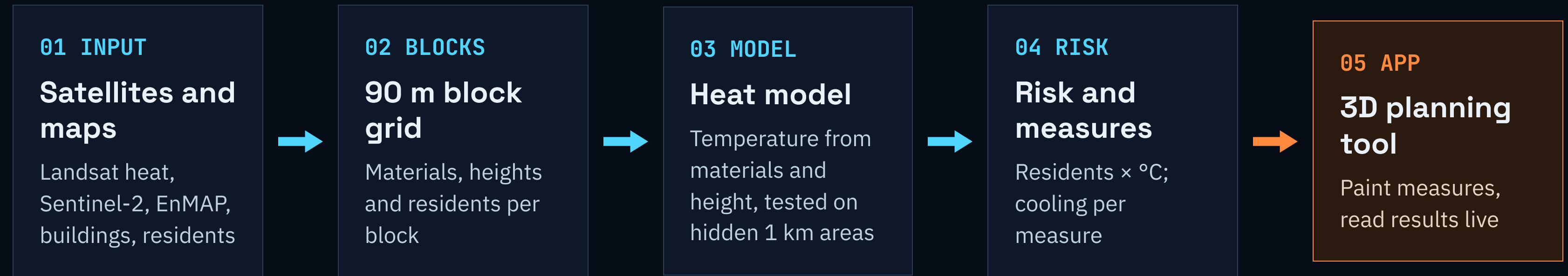
It spends the budget where it removes the most heat exposure: a third of the blocks, chosen by risk, removes 74% (slide 14).

Free satellite data that covers the whole region

Source	What we took	Used for
Landsat 8/9 (USGS)	65 summer scenes, 2023–25	Surface temperature, median per block
Sentinel-2 L2A (ESA)	70 granules (58 days), 10 m	Surface materials, vegetation
EnMAP L2A (DLR), hyperspectral	1 scene, 22 Apr 2025, 30 m	Which surfaces drive heat
OpenStreetMap, Google Open Buildings 2.5D	~1,900 buildings, heights, 2016–23	3D city, roofs, urban growth
WorldPop 2024	100 m population grid	Residents per block

Streamed from Microsoft Planetary Computer and DLR EOC Geoservice. No paid data; raw EnMAP data stays out of the repository (licence).

From satellite scenes to a planning tool



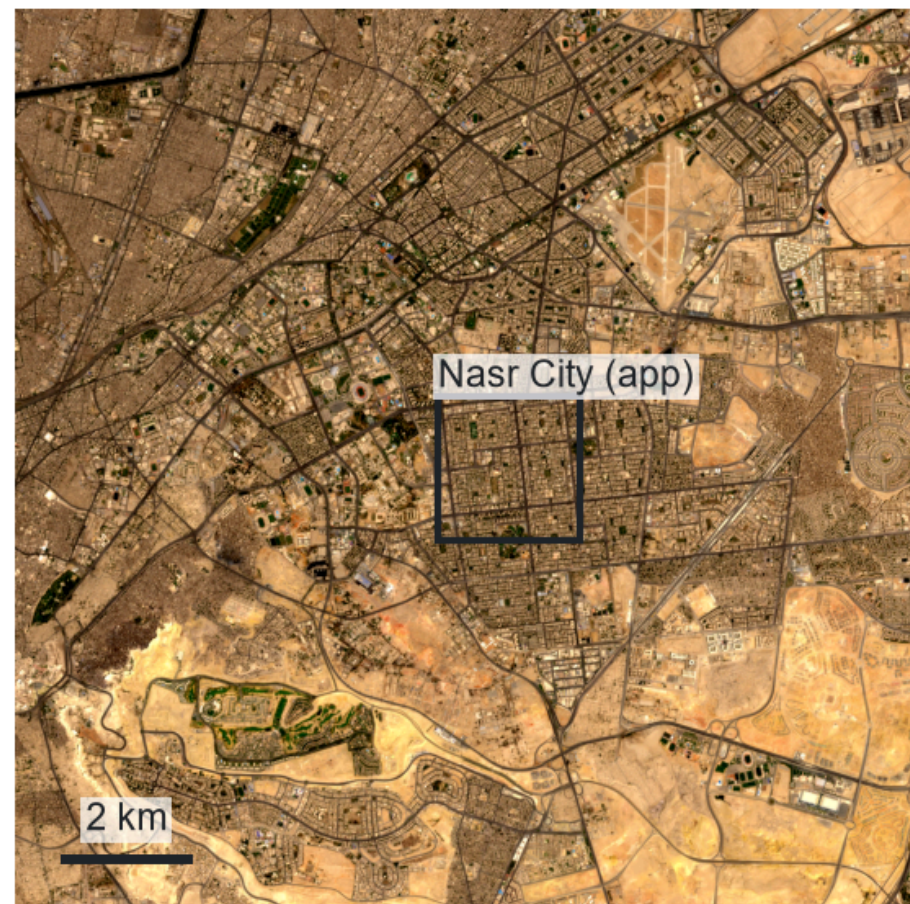
PYTHON 3.12 • uv • NOTEBOOKS 01-06 • UNIT TESTS

UNITY 6 APP

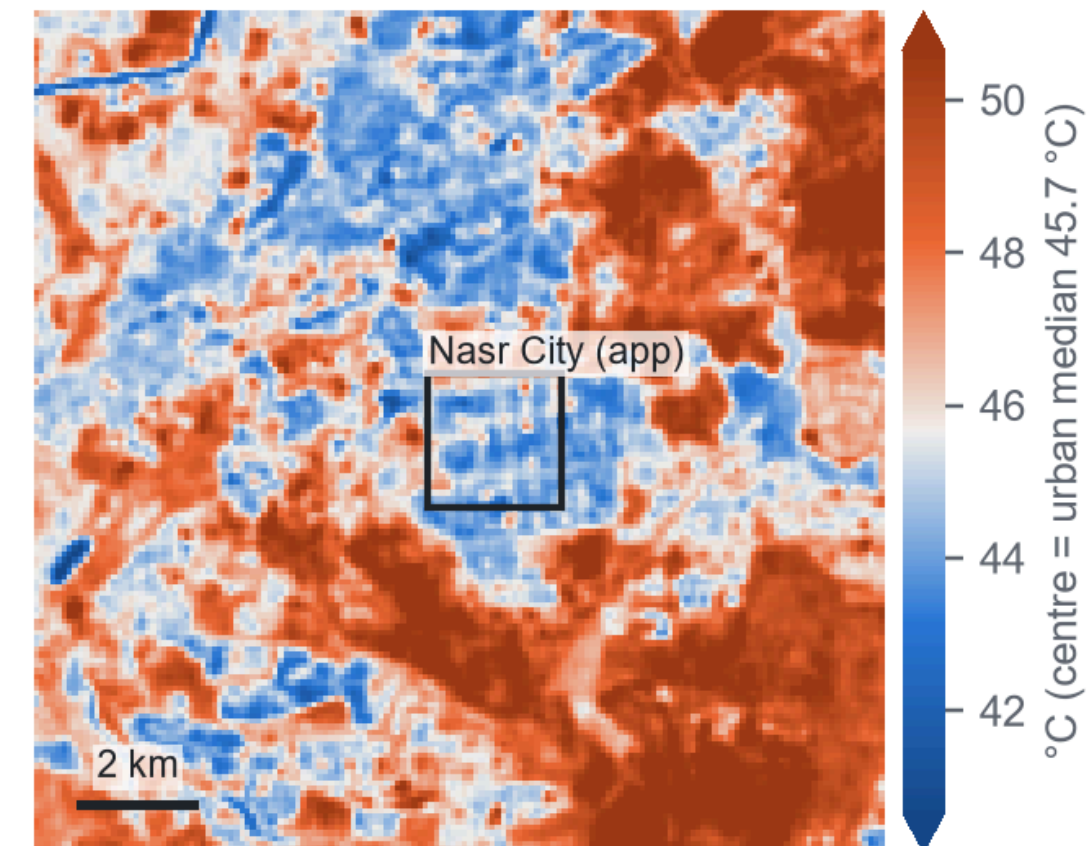
The analysis writes one file, [export/district.json](#); the app evaluates the same model live, so every number on screen traces back to a notebook.

Desert hotter than the irrigated city

a Model area, Sentinel-2 true colour



b Summer surface temperature, 90 m blocks



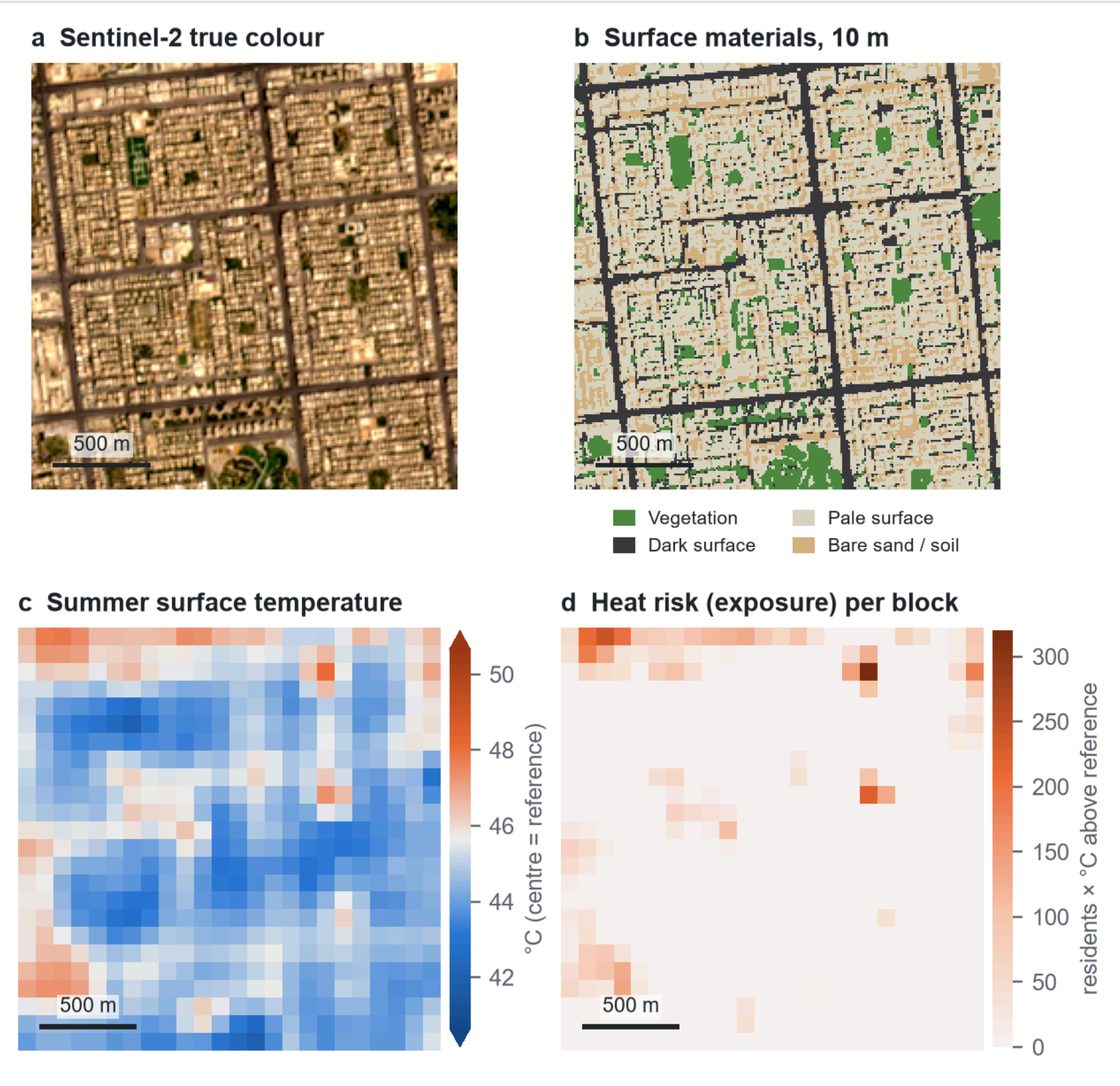
(a) Sentinel-2 summer photo; **(b)** Landsat surface temperature per 90 m block, median of 65 summer scenes, centred on the typical block (45.7 °C). Box: the Nasr City district in the app.

Heat and heat risk are not the same map

Nasr City from the analysis: (a) Sentinel-2 summer photo, (b) surface materials at 10 m, (c) summer surface temperature per 90 m block, (d) heat risk, residents × °C above the typical block.

Bare desert edges are hottest, but few people live there.

Risk concentrates where dense housing meets hot surfaces, so those blocks come first.



Hyperspectral EnMAP explains heat best

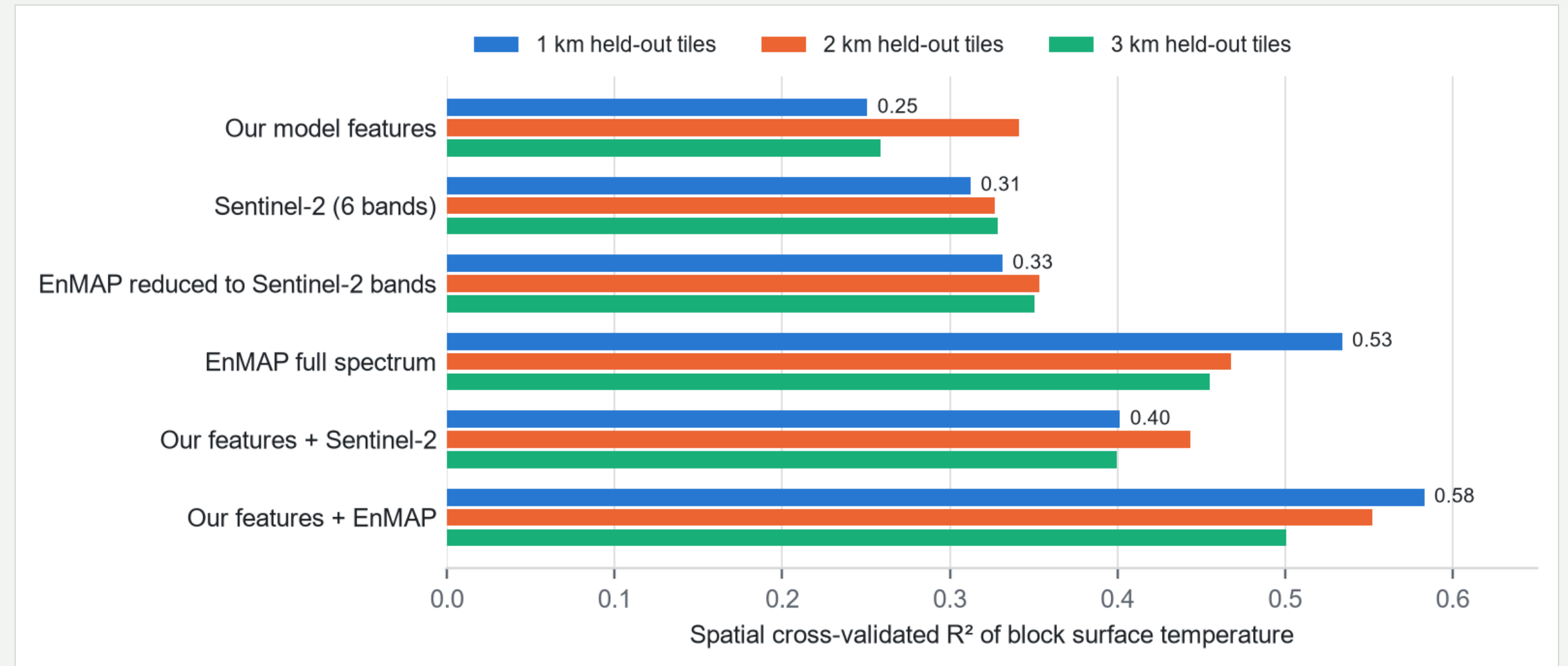
R² WITH OUR FEATURES

0.40

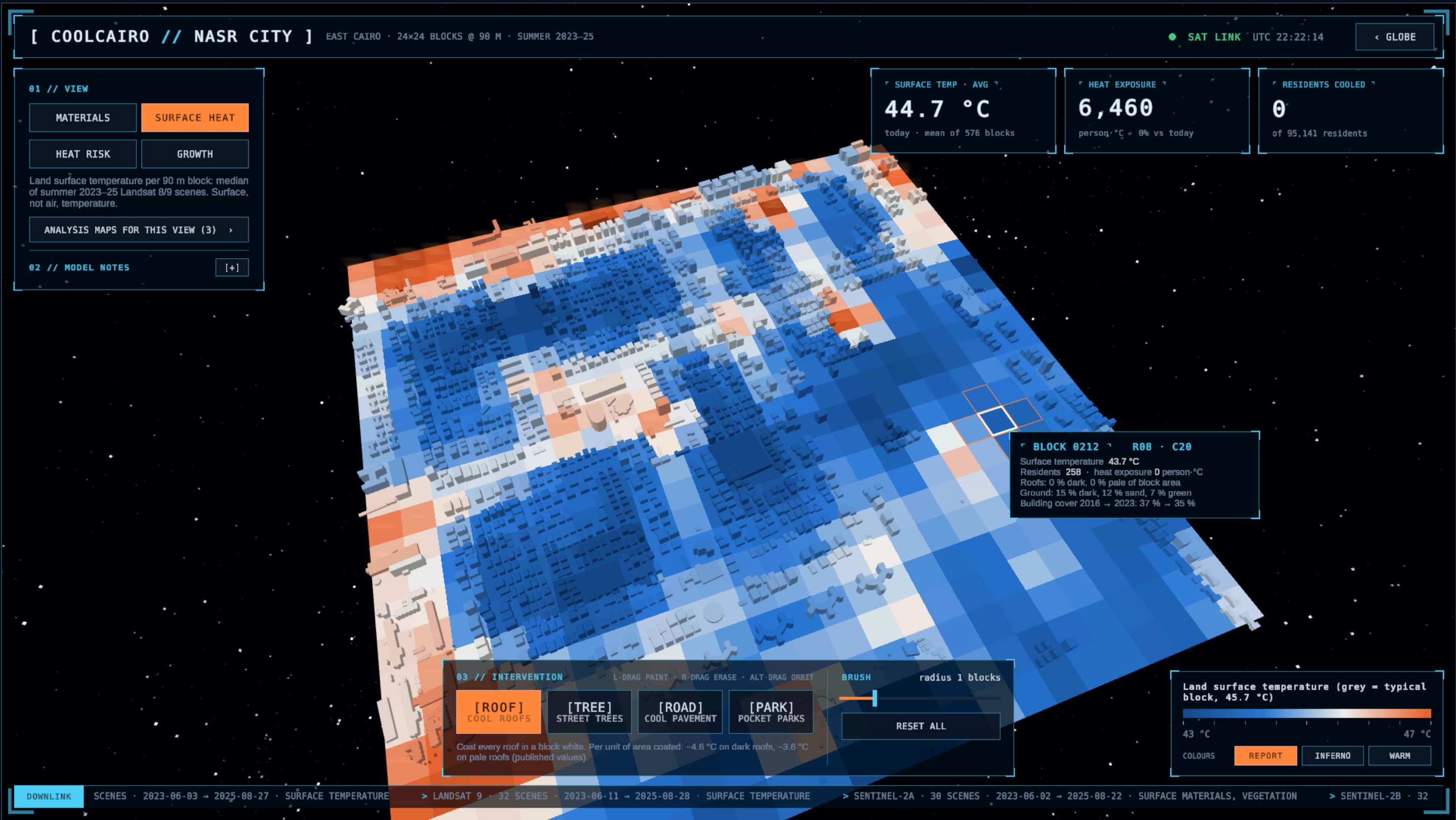
Sentinel-2 bands

0.58

EnMAP hyperspectral



Hidden 1 km areas; holds at 2 and 3 km. Spectra alone: EnMAP 0.53 vs 0.33 for the same scene cut to Sentinel-2's bands, so the gain is spectral detail. No gain for telling buildings from desert.



09 // OUTPUT · THE
TOOL, TODAY

44.7 °C

mean block surface
temperature

6,460

heat exposure, person·°C

0

of 95,141 residents in cooled
blocks

The app today: Surface heat view, summer surface temperature per 90 m block over 3D buildings. Left: views; bottom: the four cooling tools; right: the colour key.



10 // OUTPUT · THE TOOL, AFTER A PLAN

44.3 °C

mean block surface, was 44.7

1,686

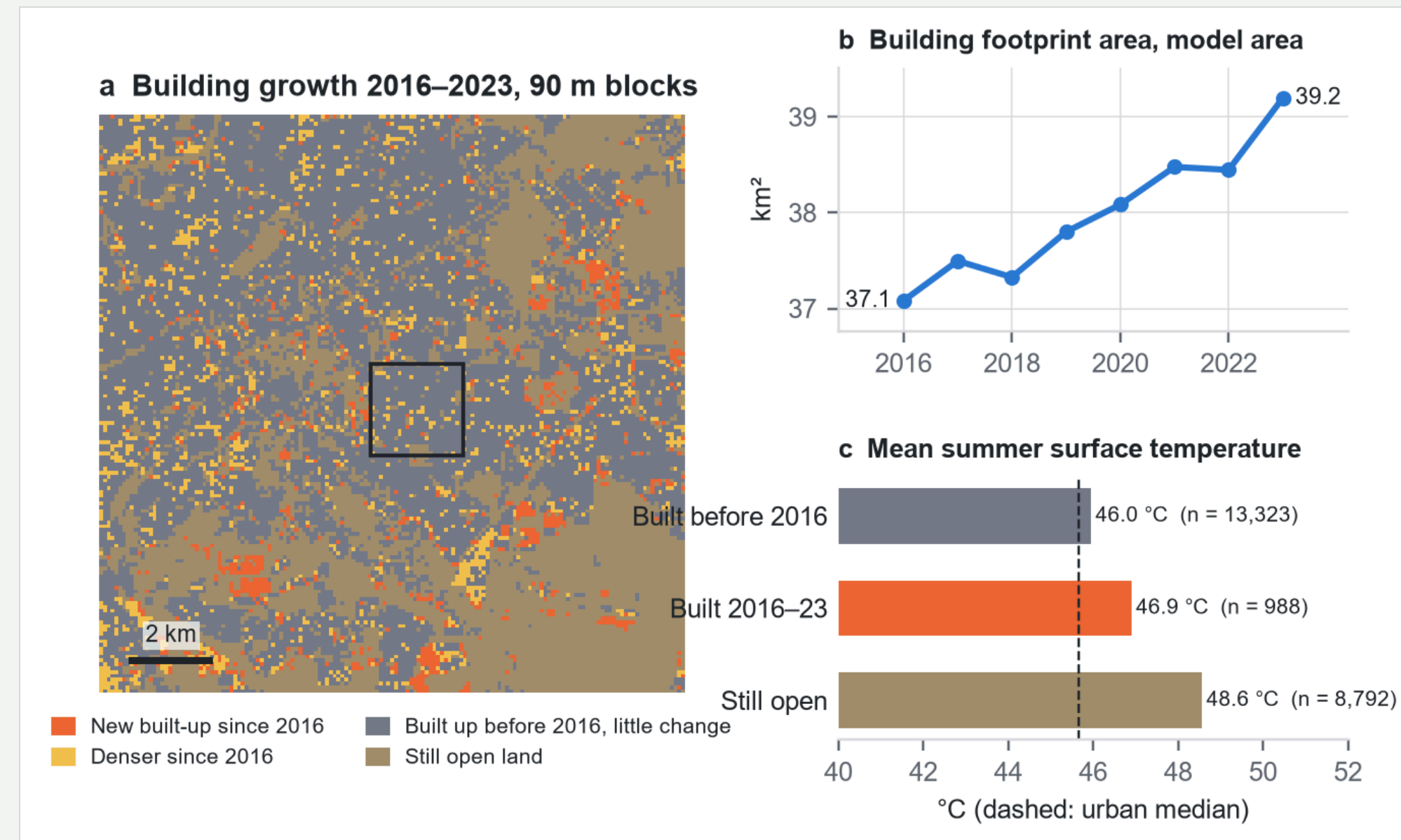
heat exposure, -74% vs today

22,588

residents in cooled blocks

After a plan: cool roofs and pocket parks painted on the riskiest third of blocks. Heat risk view: only a few blocks are still above the typical temperature.

New neighbourhoods are being built hot



+5.7%

building footprint in east Cairo, 2016–23 (37.1 → 39.2 km²)

988

blocks turned from open land to built-up, now home to ~61,000 people

46.9 °C

new blocks, vs 46.0 established and 48.6 open desert

What we checked, and what we cannot claim

R^2 0.25

Heat model on hidden 1 km areas, ± 1.2 °C, 2,538 blocks

16 / 18

material spot checks agree with satellite imagery (89%, not blind)

r 0.60

building heights vs OSM tags, bias -1.4 m, error 6.2 m

51 / 51

automated checks pass inside the built app

1 Surface, not air. Satellites measure how hot surfaces get.

3 Model explains ~a quarter. Fine for comparing measures, not exact predictions.

5 OpenStreetMap gaps. The model learns from well-mapped blocks.

7 One EnMAP scene (April). A strong signal, not yet the main model.

2 90 m blocks only. The heat sensor cannot see single buildings.

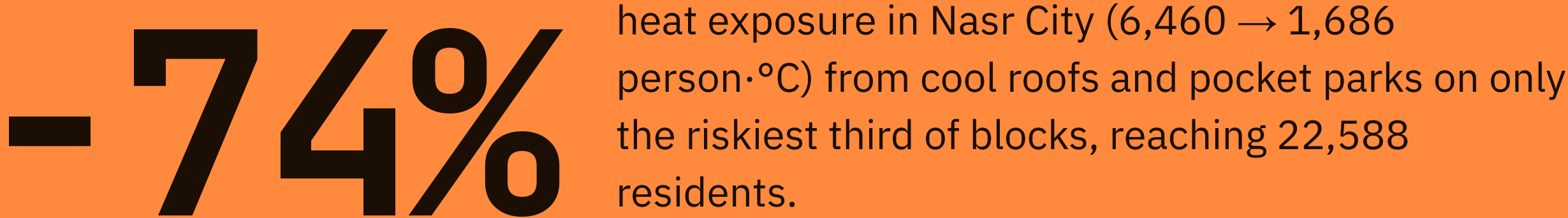
4 Dusty roofs vs sand, shadows. Why cool roofs use published physics.

6 Modelled residents. WorldPop; no age or housing vulnerability yet.

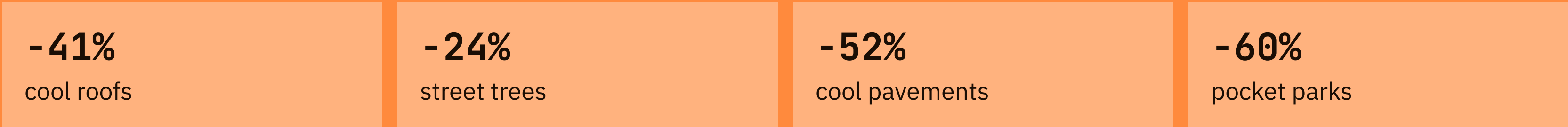
How we would overcome each limitation

Limitation	How to overcome it	Effort	Evidence it helps
1 Surface, not air temperature	Weather stations on the same days; 10–20 air sensors for one summer; SOLWEIG comfort model	1–2 days; sensors: incubation	Direct test: hot surface = hot air?
2 Block level only (90 m)	Sharpen to ~30 m with Sentinel-2; ECOSTRESS (~70 m, day and night); ESA LSTM, TRISHNA (~50 m)	Days; missions when they fly	Published; night heat matters most
3 Model explains ~25% (R ² 0.25)	Add EnMAP spectra, neighbouring blocks, desert distance, street width; test a non-linear model	Half a day	EnMAP measured: R ² 0.40 → 0.58
4 Dusty roofs vs sand; shadows	Classifier on all 6 Sentinel-2 bands; shadows from heights and sun angle; blind spot check	About 1 day	Test pixels: 76% → 86%
5 OpenStreetMap gaps in Cairo	Google Open Buildings or Microsoft footprints as the roof map; add missing buildings to OSM	1–2 days	More than today's 2,538 training blocks
6 Modelled residents; no vulnerability	WorldPop age grids (over-65s, under-5s), CAPMAS census, clinics and shade → heat vulnerability index	1–2 days; index: incubation	Standard public-health method
7 One EnMAP scene (April)	Request summer EnMAP scenes; NASA EMIT, Tanager, PRISMA; Satellite 813 data in incubation	Days to months	One scene already: 0.40 → 0.58

Targeting beats spreading: cool the right third of blocks



ONE MEASURE IN EVERY BLOCK, FOR COMPARISON



From a Nasr City prototype to a MENA heat-planning tool

Validate on the ground

10–20 air-temperature sensors in Nasr City for one summer; a blind check of the material map.

See more

Summer hyperspectral scenes (EnMAP, EMIT, Satellite 813); night-time heat from ECOSTRESS.

Map every building

Google Open Buildings and Microsoft footprints fill OpenStreetMap's gaps in Cairo.

Protect the vulnerable

Add age (over-65s, under-5s) and access to cooling; a heat vulnerability index.

Scale across MENA

Same free data and code for the next cities on our globe: Riyadh, Amman, Casablanca.

Add costs

Price per block for each measure from Egyptian supplier quotes, so planners compare °C per pound.

01 // VIEW

MATERIALS

SURFACE HEAT

HEAT RISK

GROWTH

Land surface temperature per 90 m block: median of summer 2023–25 Landsat 8/9 scenes. Surface, not air, temperature.

ANALYSIS MAPS FOR THIS VIEW (3)

02 // MODEL NOTES

[+]

SURFACE TEMP · AVG

44.7 °C

today · mean of 576 blocks

HEAT EXPOSURE

6,460

person·°C · 0% vs today

RESIDENTS COOLED

0

of 95,141 residents

[COOLCAIRO // TEAM 46 · EGYPT]

Thank you

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TEAM LEAD

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TECHNICAL LEAD

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DATA & HYPERSPECTRAL ANALYST

Code, notebooks, methodology report and the Windows app: github.com/mohamedEsmat/CoolCairo

Data: Landsat (USGS) · Contains modified Copernicus Sentinel data [2023–2025] · Contains modified EnMAP data © DLR [2025] · WorldPop 2024 (CC BY 4.0) · © OpenStreetMap contributors · Google Open Buildings 2.5D Temporal (CC BY 4.0) · NASA Blue Marble

Roofs: 0 % dark, 0 % pale of block area
Ground: 15 % dark, 12 % sand, 7 % green
Building cover 2016 → 2023: 37 % → 35 %

03 // INTERVENTION

L DRAG PAINT · R DRAG ERASE · ALT DRAG ORBIT

BRUSH

radius 1 blocks

[ROOF]

COOL ROOFS

[TREE]

STREET TREES

[ROAD]

COOL PAVEMENT

[PARK]

POCKET PARKS

Coat every roof in a block white. Per unit of area coated: −4.6 °C on dark roofs, −3.6 °C on pale roofs (published values)

RESET ALL

Land surface temperature (grey = typical block, 45.7 °C)

43 °C

47 °C

COLOURS

REPORT

INFERNO

WARM